1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7
2. Control rail and E-stop latch (no logic IC)
2a. Pi heartbeat watchdog: no heartbeat, no latch
3. Braking clamp: 56.6–56.8 V on, 55.8–55.9 V off (typ). 57.5 V worst case, into 2 x HS50 20R in parallel
4. Bus capacitors
5. 12 V converter: own LM5146 buck board (marengo_buck12) on the -X rail, pigtail J24 (block 10); DVE0125V2NBC is the drop-in fallback
5a. Brick soft start (ahead of the E-stop switch)
6. Limb ports: 1.5 mOhm shunt, own breaker (block 6a), meter, soldered pigtail pair. Pigtail fuses inline.
6a. Limb port breakers: TPS48111-Q1 + one FET per port, latch-off, enabled by RUN
7. E-stop bank A: second series bank, own driver, ahead of bank B
8. Spare 12 V outputs
8a. SPARE_ON eFuse
9. E-stop pressed: button NO contact to the Pi
10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)
10b. Absorber_close option (M3): DR fitted, the rest DNP
10c. Rev B4 review parts
10d. Rev B5 parts
TP7
TP GND
TP8
TP GND
TP9
TP GND
TP10
TP GATE
TP11
TP GATE_A
TP12
TP UVLO
TP13
TP FLT_A

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR

marengo

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Title: marengo power board rev B

Size: A0 | Date: 2026-11-06 | **Rev: B-wip**
KiCad E.D.A. 10.0.8 | Id: 1/1

Fifth 47 uF 80 V hybrid on VBUS (EEH-ZC1K470P, active line; the APZ 68 uF is obsolete): 235 uF in all, the value tools/port_sim.py uses.